INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I , Fourth Semester Examination, February- 2025

Subject Code: BCACC403

Subject Name: Operating Systems

Full Marks: 30 Set No: 1 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Define an Operating System. List its key functions. **2**

2. Discuss the objectives of an Operating System. **2**

3. What is a process? **2**

4. Why is Thread known as a light weight process? **2**

5. Define Job Queue. **2**

6. What does PCB contain? **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Discuss about Orphan Process. **5**

8. Describe Process Control Block. **5**

9. Explain the property of fork () function. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Illustrate about the concept of Context switching in Process? How does it take place and what is the importance of Context Switching. **10**

11. Consider the following table and evaluate the average turnaround time and average waiting time for SJF. 10

Process Name	Arrival Time	Processing Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

--End--

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Full Marks: 30 Set No: 2 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. What is the difference between a program and a process? 2

2. What is Kernel? 2

3. What is the role of a process scheduler? 2

4. Differentiate between process and thread. 2

5. What is the role of a dispatcher in CPU scheduling? 2

6. What is preemptive and non-preemptive scheduling? 2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Explain the different states of a process with a suitable diagram. 5

8. Compare and contrast multitasking, multiprocessing, and multithreading. 5

9. Discuss the difference between short-term, medium-term, and long-term scheduling? 5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Describe different types of operating systems (Batch, Time-Sharing, Distributed, Real-Time, and Embedded) with their advantages, disadvantages, and real-world applications 10

11. Consider the following table and evaluate the average turnaround time and average waiting time for SRTF scheduling algorithm. 10

Process Name	CPU Brust	Arrival Time
P1	2	7
P2	5	3
P3	2	4
P4	9	0
P5	2	15

--End--

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Full Marks: 30 Set No: 3 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. What is Mode bit? 2

2. Define Throughput. 2

3. Define Response Time. 2

4. What are PID and PPID? 2

5. What is a real-time operating system (RTOS)? 2

6. Give two examples of open-source operating systems. 2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Discuss about init process. 5

8. Describe the concept of inter-process communication (IPC) and its mechanisms. 5

9. What is a context switch? Explain the steps involved in a context switch. 5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Write the Algorithm for FCFS and SRJF. Also write the advantages and disadvantages of both the algorithms. 10

11. Consider the following table and evaluate the average turnaround time and average waiting time for FCFS. 10

Process Name	Arrival Time	Processing Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

--End--